MATH60005/70005: Optimisation Coursework (AY 2627)

Instructions: read this first![1]

- This coursework has a total of 20 marks and accounts for 10% of the module.
- **Submission deadline:** Monday, 23 November, 13:00 UK time, via Blackboard dropbox.
- **Submit a single PDF file typeset in LaTeX, Word, or similar.**
- The coursework requires basic coding capabilities that are available in all modern languages. **You are welcome to use any language of your preference**. You don't need to include your code in the submission, we want to assess your answers/plots/mathematical reasoning.
- **Marking criteria:** Full marks will be awarded for work that 1) is mathematically correct, 2) shows an understanding of material presented in lectures, 3) gives details of all calculations, implementation, and reasoning, and 4) is presented in a logical and clear manner.
- Do not discuss your answers publicly via our forum. If you have any queries regarding your interpretation of the questions, please contact the lecturer at `dkaliseb@imperial.ac.uk`. Consulting with the lecturer excludes debugging code or checking whether an answer is correct.
- Beware of plagiarism regulations; the College takes this very seriously. This is a **group-based assessment** with groups of 1 to 3 students. **Make a single group submission displaying the CID of every group member on the front page. Do not include your name.**

Part I: Gradient-Enhanced Polynomial Regression

Consider the task of approximating a function $V : \mathbb{R}^d \to \mathbb{R}$ from a set of m random samples $\mathcal{D} = \{\mathbf{x}_i, V(\mathbf{x}_i)\}_{i=1}^m$ via linear least squares approximation. We will fit polynomial models of the form

$$V_\theta(\mathbf{x}) = \sum_{j=1}^{n} \theta_j \phi_j(\mathbf{x}) = \theta^\top \Phi(\mathbf{x}) \,,$$

where $\theta = (\theta_1, \ldots, \theta_n)^\top$ are the expansion coefficients to be learned, and $\Phi(\mathbf{x}) = (\phi_1(\mathbf{x}), \ldots, \phi_n(\mathbf{x}))^\top$ is a known set of polynomial basis functions in $\mathbb{R}^d$ to be specified later.

I.a) **[3 marks]** Formulate the approximation problem above as a linear least squares of the form

$$\min_{\theta \in \mathbb{R}^n} \|\mathbf{A}\theta - \mathbf{b}\|_2^2 \,,$$

providing general but precise expressions for $\mathbf{A}$ and $\mathbf{b}$.

[1] The ability to read and follow instructions is a highly valued skill in the job market and elsewhere.

Using your previous formulation, fit a model for the 1-d function data given in `trainingIa.dat`[2]. Here, the set $\Phi(x) = (1, x, x^2, \ldots, x^{n-1})$ corresponds to the 1-d monomial basis up to degree $n-1$. Present a plot displaying MSE $:= \frac{1}{m_{val}} \sum_{k=1}^{m_{val}} |V_\theta(x_k) - V(x_k)|^2$ in the validation data versus degree n, with $n = 1, \ldots, 20$. Use all the available training data, and measure your validation errors using `validationIa.dat`. What is the smallest n for which you achieve an error of 10^{-3} on the validation set? Fixing the value of n, plot the MSE in validation versus number of training points. Show the function that is learned at that value of n.

I.b) **[4 marks]** Now, assume that your training dataset has been enriched with derivative information, $\mathcal{D} = \{\mathbf{x}_i, V(\mathbf{x}_i), \nabla V(\mathbf{x}_i)\}_{i=1}^m$, available in the file `trainingIb.dat`. Using the same model $V_\theta(\mathbf{x})$ and monomial basis as in part I.a), reformulate the linear least squares problem to incorporate this data. Repeat the plots and analysis, and compare both regressions using the validation dataset `validationIa.dat`. What do you observe?

Part II: A First Approach to Dynamic Optimisation

Consider a discrete dynamical process of the form

$$\begin{aligned} &x_0 = \bar{x} \in \mathbb{R}\,, \\ &x_i = a x_{i-1} + b u_i\,, \quad i = 1, \ldots, N\,, \end{aligned} \tag{S}$$

where $a, b \in \mathbb{R}$. The variables x_i and u_i denote the internal state of the system and a control variable at discrete time i, respectively. The sequence $\mathbf{x}_{\bar{x}}^{\mathbf{u}} := \{x_i\}_{i=0}^N \in \mathbb{R}^{N+1}$ is the trajectory of the system departing from the initial condition $\bar{x}$ associated with the control sequence $\mathbf{u} := \{u_i\}_{i=1}^N \in \mathbb{R}^N$.

Given an initial condition $\bar{x}$, and parameters a, b, and N, our goal is to find an optimal sequence of controls $\mathbf{u}$ which drives the trajectory of the system $\mathbf{x}_{\bar{x}}^{\mathbf{u}}$ to zero while balancing the amount of control energy spent on this task. We express our goal as the **dynamic optimisation problem**

$$\min_{\mathbf{u} \in \mathbb{R}^N} \ \|\mathbf{x}_{\bar{x}}^{\mathbf{u}}\|_2^2 + \gamma \|\mathbf{u}\|_2^2\,, \qquad \gamma > 0\,, \tag{DO}$$

subject to

$$\begin{aligned} &x_0 = \bar{x}\,, \\ &x_i = a x_{i-1} + b u_i\,, \quad i = 1, \ldots, N. \end{aligned}$$

II.a) **[3 marks]** Reformulate (DO) as a regularised linear least squares problem for $\mathbf{u}$. Discuss the existence and uniqueness of an optimal solution $\mathbf{u}^*$ to this problem. Show that any $\mathbf{u}^*$ solving the associated regularised problem satisfies $\|\mathbf{u}^*\| \leq \|\mathbf{u}\|$ for any $\mathbf{u}$ solving the unregularised linear least squares problem (that is, with $\gamma = 0$).

For $N = 50$, $a = 1$, $b = -0.01$, and $\bar{x} = 1$, generate two plots: one for the optimal control signals $\mathbf{u}^*$, and another for the associated optimal trajectories $\mathbf{x}_{\bar{x}}^{\mathbf{u}^*}$, for $\gamma = 10^{-3}, 10^{-2}, 0.1, 1$. What is the effect of increasing γ on both the control and the trajectories?

[2] Data format is $(x_i, V(x_i))$

II.b) **[2 marks]** Now, in addition to the system (S), consider a second system (S') given by

$$\begin{aligned} y_0 &= \bar{y} \in \mathbb{R}\,, \\ y_i &= cy_{i-1} + dv_i\,, \quad i = 1, \ldots, N\,, \end{aligned} \tag{S'}$$

and consider the problem

$$\min_{(\mathbf{u},\mathbf{v})\in\mathbb{R}^{2N}} \ \|\mathbf{x}_{\bar{x}}^{\mathbf{u}}\|_2^2 + \|\mathbf{y}_{\bar{y}}^{\mathbf{v}}\|_2^2 + \gamma\|\mathbf{u}-\mathbf{v}\|_2^2\,, \qquad \gamma > 0\,, \tag{MO}$$

that is, we want to drive both $\mathbf{x}$ and $\mathbf{y}$ to zero with the respective control signals $\mathbf{u}$ and $\mathbf{v}$ close to each other. Formulate (MO) as a regularised linear least squares problem. For $N = 50$, $a = 1$, $b = 0.05$, $\bar{x} = 1$, $c = 0.2$, $d = -0.5$, and $\bar{y} = 1$, solve the regularised linear least squares problem for $\gamma = 1$ and show two plots for state trajectories and controls.

II.c) **[4 marks]** Using the regularised least squares formulation from the previous part, if every parameter of the problem is fixed except for γ, we consider the following functionals evaluated at the corresponding optimal solution:

$$\mathcal{J}_1(\gamma) := \|\mathbf{x}^*(\gamma)\|_2^2 + \|\mathbf{y}^*(\gamma)\|_2^2\,, \qquad \mathcal{J}_2(\gamma) := \|\mathbf{u}^*(\gamma) - \mathbf{v}^*(\gamma)\|_2^2\,.$$

Using the same parameters as in part II.b), solve the regularised least squares problem for values of $10^{-5} \le \gamma \le 10^5$ (take the exponent increasing in steps of 0.1). Show a plot of $\mathcal{J}_1(\gamma)$ and $\mathcal{J}_2(\gamma)$ versus γ. Plot the curve $(\mathcal{J}_1(\gamma), \mathcal{J}_2(\gamma))$. This curve is known as the *Pareto front* and is a relevant object in multi-objective optimisation. Draw a conclusion from these plots regarding the impact of γ on the total cost.

II.d) **[4 marks]** We wish to promote sparsity in the control penalty by considering an ℓ_1 norm penalty in the cost:

$$\min_{(\mathbf{u},\mathbf{v})\in\mathbb{R}^{2N}} \ \|\mathbf{x}_{\bar{x}}^{\mathbf{u}}\|_2^2 + \|\mathbf{y}_{\bar{y}}^{\mathbf{v}}\|_2^2 + \gamma_2\|\mathbf{u}-\mathbf{v}\|_2^2 + \gamma_1\|\mathbf{u}-\mathbf{v}\|_1\,, \qquad \gamma_1, \gamma_2 \ge 0\,.$$

However, we have not yet discussed how to deal with the non-differentiability of the ℓ_1 norm at the origin (we will do so by the end of term!). Instead, we propose the following approximation to the ℓ_1 norm ($\epsilon > 0$):

$$\|\mathbf{u}\|_1 \approx \sum_{i=1}^{N} \mathcal{L}_\epsilon(u_i)\,, \qquad \mathcal{L}_\epsilon(u_i) := \begin{cases} \frac{1}{2}u_i^2 & \text{if } \ |u_i| \le \epsilon \\ \epsilon\left(|u_i| - \frac{1}{2}\epsilon\right) & \text{otherwise.} \end{cases}$$

Explain in your own words (you may use plots) the meaning of $\mathcal{L}_\epsilon(u_i)$ as a regulariser. Is it a differentiable function?

Implement a gradient descent method with constant stepsize (describe all your settings) to find the optimal solution to

$$\min_{(\mathbf{u},\mathbf{v})\in\mathbb{R}^{2N}} \ \|\mathbf{x}_{\bar{x}}^{\mathbf{u}}\|_2^2 + \|\mathbf{y}_{\bar{y}}^{\mathbf{v}}\|_2^2 + \gamma_2\|\mathbf{u}-\mathbf{v}\|_2^2 + \gamma_1\sum_{i=1}^{N}\mathcal{L}_\epsilon(u_i - v_i)\,, \qquad \gamma_1, \gamma_2, \epsilon > 0\,.$$

Using the same parameters as in part II.b), compare trajectory and control plots for

i) $\epsilon = 1$, $\gamma_2 = 1$, $\gamma_1 = 0$ (same as part II.b));

ii) $\epsilon = 1$, $\gamma_2 = 0$, $\gamma_1 = 1$;

iii) $\epsilon = 0.1$, $\gamma_2 = 0$, $\gamma_1 = 1$.

What do you observe?

Marking Scheme

Question	Marks	Assessment Criteria
I.a	3	**Formulation(1 mark):** formulation of least squares problem with precise matrix expressions for A and b. **Implementation (2 marks):** plots of MSE vs degree and MSE vs training size, identification of smallest n achieving MSE $< 10^{-3}$. Clear explanation, well-labeled figures, learned function plot.
I.b	4	**Formulation (2 marks):** reformulation incorporating gradient data with augmented system. **Implementation (2 marks):** plots and analysis, meaningful comparison with Part I.a. Clear observations about gradient-enhanced regression benefits.
II.a	3	**Theory (2 marks):** reformulation as regularized least squares, existence/uniqueness discussion, proof of $\|u^*\| \leq \|u\|$. **Implementation (1 mark):** plots for controls and trajectories with different γ values. Clear explanation of γ effect.
II.b	2	**Formulation (1 mark):** formulation of multi-objective problem as regularized least squares. **Implementation (1 mark):** solution and plots for $\gamma = 1$.
II.c	4	**Implementation (2 marks):** computation of $J_1(\gamma)$ and $J_2(\gamma)$ over specified range. **Plots (2 marks):** plots of functionals vs γ and Pareto front. Relevant conclusions about trade-offs in multi-objective optimization.
II.d	4	**Theory (1 marks):** Clear explanation of L_ϵ as regularizer and differentiability analysis. **Implementation (3 marks):** Correct gradient descent implementation with documented settings, convergence for all three cases. Meaningful comparison and observations about sparsity promotion.
Total	**20**	

Part I: Gradient-Enhanced Polynomial Regression

Part I.a [3 marks]

We seek to approximate $V : \mathbb{R}^d \to \mathbb{R}$ using a polynomial model:

$$V_\theta(x) = \sum_{j=1}^{n} \theta_j \phi_j(x) = \theta^\top \Phi(x).$$

Given training data $\mathcal{D} = \{x_i, V(x_i)\}_{i=1}^m$, we want to find $\theta \in \mathbb{R}^n$ that minimizes the squared error:

$$\min_{\theta \in \mathbb{R}^n} \sum_{i=1}^{m} |V_\theta(x_i) - V(x_i)|^2.$$

This can be formulated as a linear least squares problem:

$$\min_{\theta \in \mathbb{R}^n} \|A\theta - b\|_2^2$$

where $A \in \mathbb{R}^{m \times n}$ with entries $A_{ij} = \phi_j(x_i)$ for $i = 1, \ldots, m$ and $j = 1, \ldots, n$, and $b \in \mathbb{R}^m$ with entries $b_i = V(x_i)$ for $i = 1, \ldots, m$. For the 1-d monomial basis $\Phi(x) = (1, x, x^2, \ldots, x^{n-1})^\top$, we have:

$$A_{ij} = x_i^{j-1}, \quad i = 1, \ldots, m, \quad j = 1, \ldots, n,$$

and the optimal solution is obtained by solving the normal equations:

$$A^\top A\theta = A^\top b.$$

Numerical Results

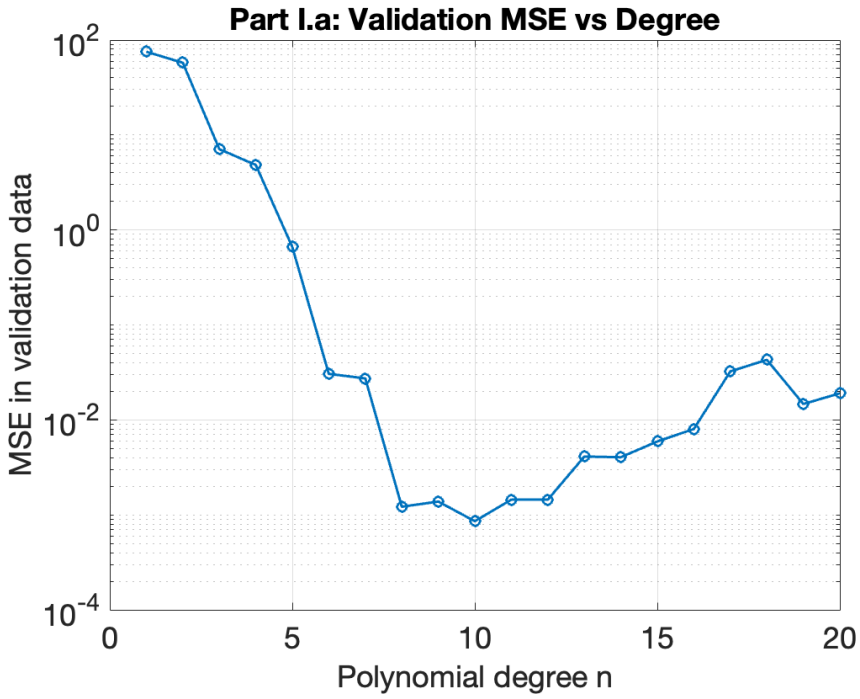


Figure 1: Validation MSE versus polynomial degree for standard regression. The smallest degree n achieving MSE $< 10^{-3}$ is $n = 10$.

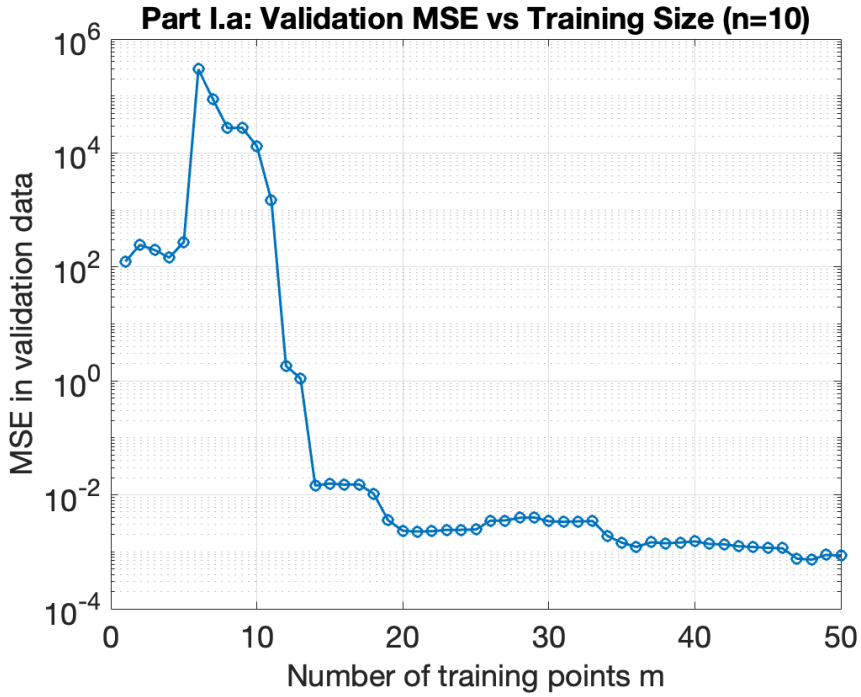


Figure 2: Validation MSE versus number of training points with fixed polynomial degree $n = 10$. The error decreases as more training data becomes available.

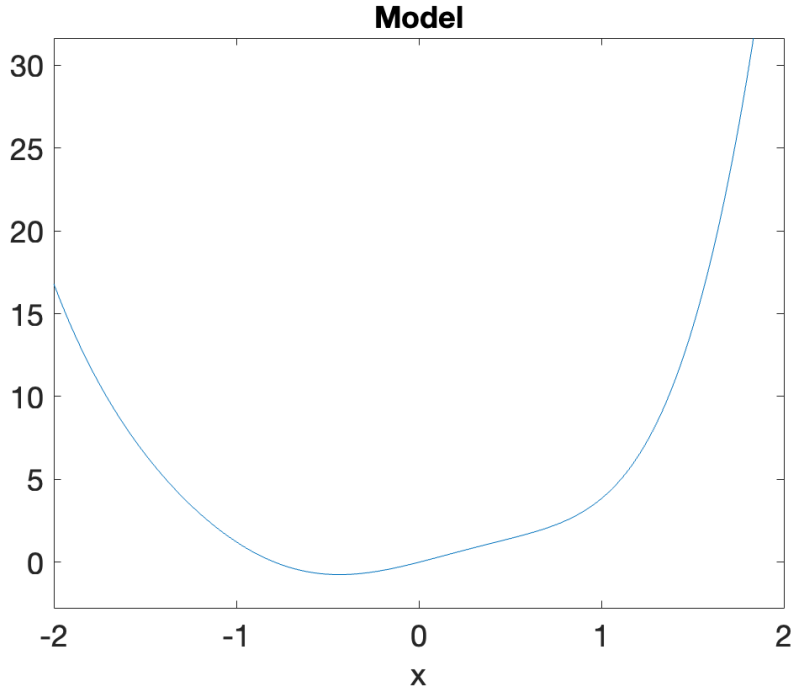


Figure 3: The learned polynomial function $V_\theta(x)$ with degree $n = 10$ over the interval $[-2, 2]$.

Part I.b [4 marks]

Now we have enriched training data: $\mathcal{D} = \{x_i, V(x_i), \nabla V(x_i)\}_{i=1}^m$. Using the same polynomial model $V_\theta(x) = \sum_{j=1}^n \theta_j \phi_j(x)$, the gradient is:

$$\nabla V_\theta(x) = \sum_{j=1}^n \theta_j \nabla \phi_j(x).$$

For 1-d monomials: $\phi_j(x) = x^{j-1}$, so $\frac{d\phi_j}{dx}(x) = (j-1)x^{j-2}$. We minimize both function and gradient errors:

$$\min_{\theta \in \mathbb{R}^n} \left[\sum_{i=1}^m |V_\theta(x_i) - V(x_i)|^2 + \sum_{i=1}^m |\nabla V_\theta(x_i) - \nabla V(x_i)|^2 \right],$$

And this is reformulated as:

$$\min_{\theta \in \mathbb{R}^n} \|A\theta - b\|_2^2,$$

where $A \in \mathbb{R}^{2m \times n}$ and $b \in \mathbb{R}^{2m}$ are given by

$$A = \begin{bmatrix} x_1^0 & x_1^1 & \cdots & x_1^{n-1} \\ \vdots & \vdots & \ddots & \vdots \\ x_m^0 & x_m^1 & \cdots & x_m^{n-1} \\ 0 & 1 \cdot x_1^0 & \cdots & (n-1)x_1^{n-2} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 1 \cdot x_m^0 & \cdots & (n-1)x_m^{n-2} \end{bmatrix}, \qquad b = \begin{bmatrix} V(x_1) \\ \vdots \\ V(x_m) \\ \nabla V(x_1) \\ \vdots \\ \nabla V(x_m) \end{bmatrix}.$$

Numerical Results

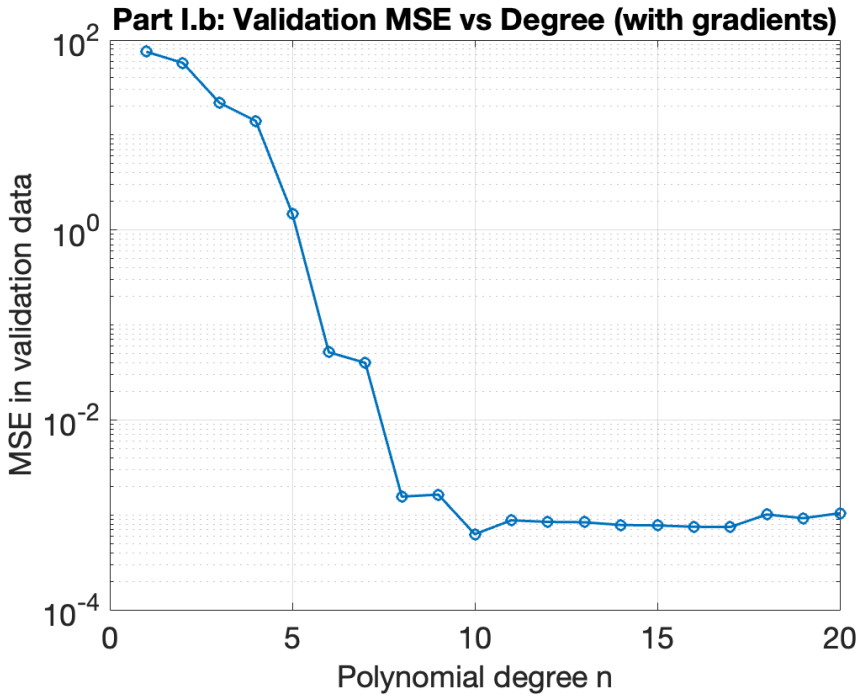


Figure 4: Validation MSE versus polynomial degree for gradient-enhanced regression. Similarly to the previous regression, the smallest degree n achieving MSE $< 10^{-3}$ is $n = 10$. However, we don't observe a dramatic overfitting when increasing the polynomial degree.

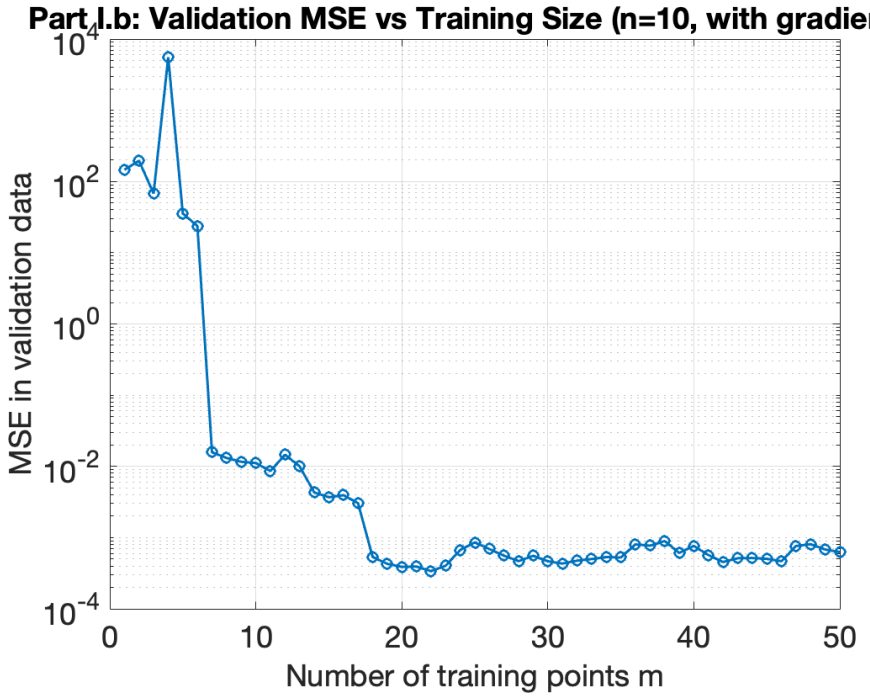


Figure 5: Validation MSE versus number of training points with gradient information. We reach a consistent validation error of 10^{-3} with fewer training points compared to the regular regression.

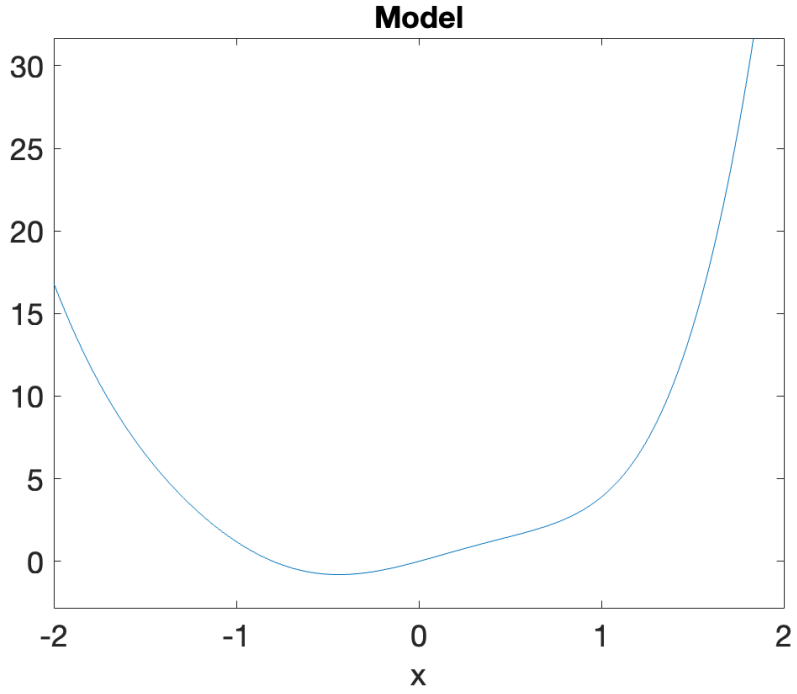


Figure 6: The learned polynomial function with gradient-enhanced regression.

Part II: A First Approach to Dynamic Optimisation

Part II.a [3 marks]

Consider the discrete dynamical system:

$$\begin{aligned} x_0 &= \bar{x} \\ x_i &= ax_{i-1} + bu_i, \quad i = 1, \dots, N \end{aligned}$$

We can express the trajectory $x = \{x_i\}_{i=0}^N$ as a linear function of the control $u = \{u_i\}_{i=1}^N$. By iterating the dynamics:

$$\begin{aligned} x_1 &= ax_0 + bu_1 = a\bar{x} + bu_1, \\ x_2 &= ax_1 + bu_2 = a^2\bar{x} + abu_1 + bu_2, \\ x_3 &= ax_2 + bu_3 = a^3\bar{x} + a^2bu_1 + abu_2 + bu_3, \\ x_i &= a^i\bar{x} + \sum_{j=1}^{i} a^{i-j}bu_j, \end{aligned}$$

can write $x = Bu + c$ where: $x \in \mathbb{R}^{N+1}$ is the state trajectory, $u \in \mathbb{R}^N$ is the control sequence, $B \in \mathbb{R}^{(N+1)\times N}$ with

$$B_{i,j} = \begin{cases} ba^{i-j-1} & \text{if } j \leq i-1 \\ 0 & \text{otherwise} \end{cases}$$

for $i = 1, \dots, N$, $B_{0,j} = 0$, and $c \in \mathbb{R}^{N+1}$ with $c_0 = \bar{x}$ and $c_i = a^i\bar{x}$ for $i = 1, \dots, N$. Then, the dynamic optimization problem (DO) becomes a regularized least squares problem of the form:

$$\min_{u \in \mathbb{R}^N} \|Bu + c\|_2^2 + \gamma\|u\|_2^2.$$

The objective function is strictly convex and a coercive (the objective is a positive definite quadratic form), and hence there exists a unique global minimizer u^* given by

$$u^* = -(B^\top B + \gamma I)^{-1}B^\top c.$$

We denote by u the solution to the unregularized problem (i.e., $\gamma = 0$):

$$\min_{u \in \mathbb{R}^N} \|Bu + c\|_2^2$$

By optimality of u^* for the regularized problem, we have

$$\|Bu^* + c\|_2^2 + \gamma\|u^*\|_2^2 \leq \|Bu + c\|_2^2 + \gamma\|u\|_2^2.$$

However, the optimality of u for the unregularized problem yields

$$\|Bu + c\|_2^2 \leq \|Bu^* + c\|_2^2$$

together with

$$\|Bu^* + c\|_2^2 + \gamma\|u^*\|_2^2 \leq \|Bu^* + c\|_2^2 + \gamma\|u\|_2^2,$$

which implies

$$\gamma\|u^*\|_2^2 \leq \gamma\|u\|_2^2.$$

Numerical Results

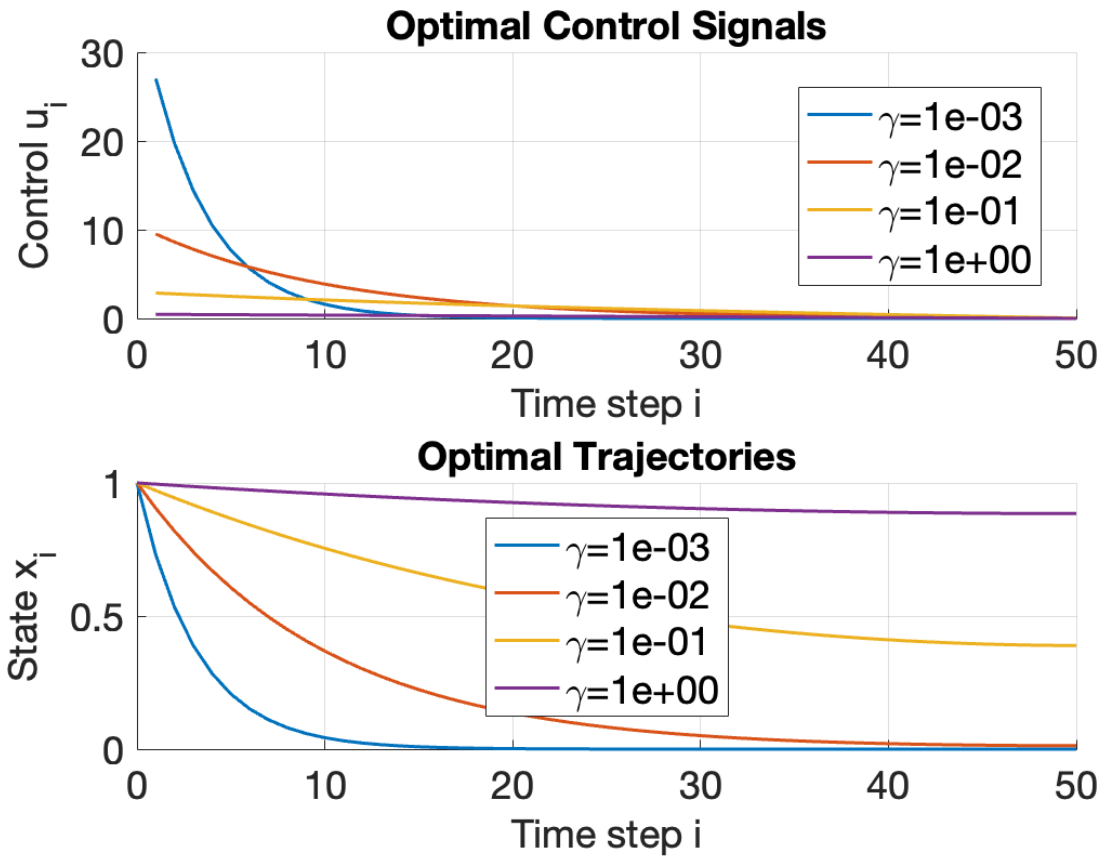


Figure 7: Optimal control signals (top) and state trajectories (bottom) for different values of γ. Larger γ values penalize the control effort, leading to smaller $\|u^*\|$, and less effective reduction of $\|x^*\|$.

Part II.b [2 marks]

We now have two systems:

$$\begin{aligned} \text{System } (S): \quad & x_0 = \bar{x}, \quad x_i = ax_{i-1} + bu_i \\ \text{System } (S'): \quad & y_0 = \bar{y}, \quad y_i = cy_{i-1} + dv_i \end{aligned}$$

The multi-objective problem (MO) is:

$$\min_{(u,v)\in\mathbb{R}^{2N}} \|x\|_2^2 + \|y\|_2^2 + \gamma\|u - v\|_2^2$$

Following the same approach as Part II.a, we have:

$$\begin{aligned} x &= B_x u + c_x \\ y &= B_y v + c_y \end{aligned}$$

where B_x, c_x are constructed from parameters $(a, b, \bar{x})$ and B_y, c_y from $(c, d, \bar{y})$.
The problem becomes:

$$\min_{(u,v)\in\mathbb{R}^{2N}} \|B_x u + c_x\|_2^2 + \|B_y v + c_y\|_2^2 + \gamma\|u - v\|_2^2$$

Define $w = \begin{bmatrix} u \\ v \end{bmatrix} \in \mathbb{R}^{2N}$. We can write:

$$\min_{w\in\mathbb{R}^{2N}} \|Aw + b\|_2^2 + \gamma\|Cw\|_2^2,$$

where:

$$A = \begin{bmatrix} B_x & 0 \\ 0 & B_y \end{bmatrix} \in \mathbb{R}^{2(N+1)\times 2N}, \quad b = \begin{bmatrix} c_x \\ c_y \end{bmatrix} \in \mathbb{R}^{2(N+1)}, \quad C = \begin{bmatrix} I_N & -I_N \end{bmatrix} \in \mathbb{R}^{N\times 2N}.$$

This is a regularized least squares problem. The optimal solution satisfies:

$$w^* = -(A^\top A + \gamma C^\top C)^{-1} A^\top b.$$

Numerical Results

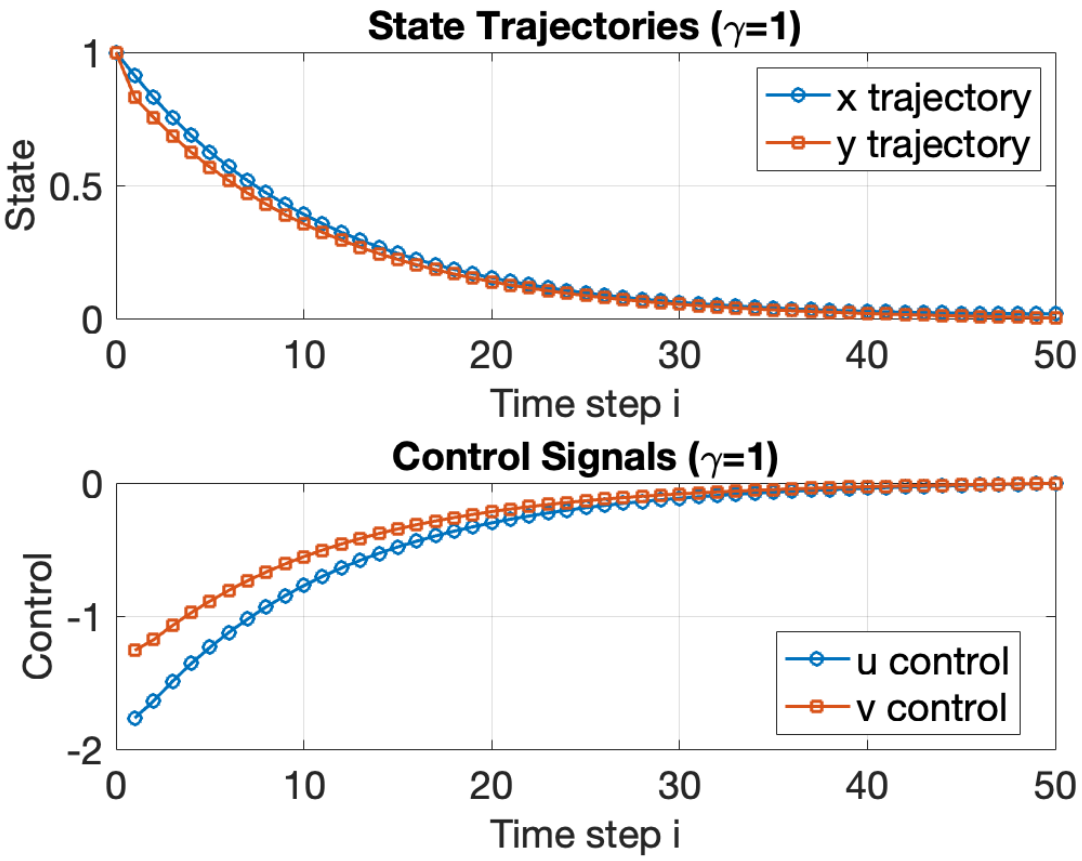


Figure 8: State trajectories (top) and control signals (bottom) for the multi-objective problem with $\gamma = 1$. The penalty term effectively promotes the control signals being similar, while achieving stabilization for both trajectories (x, y).

Part II.c [4 marks]

For varying $\gamma \in [10^{-5}, 10^{5}]$, we compute:

$$J_1(\gamma) = \|x^*(\gamma)\|_2^2 + \|y^*(\gamma)\|_2^2 \quad \text{(trajectory error)}$$
$$J_2(\gamma) = \|u^*(\gamma) - v^*(\gamma)\|_2^2 \quad \text{(control similarity)}$$

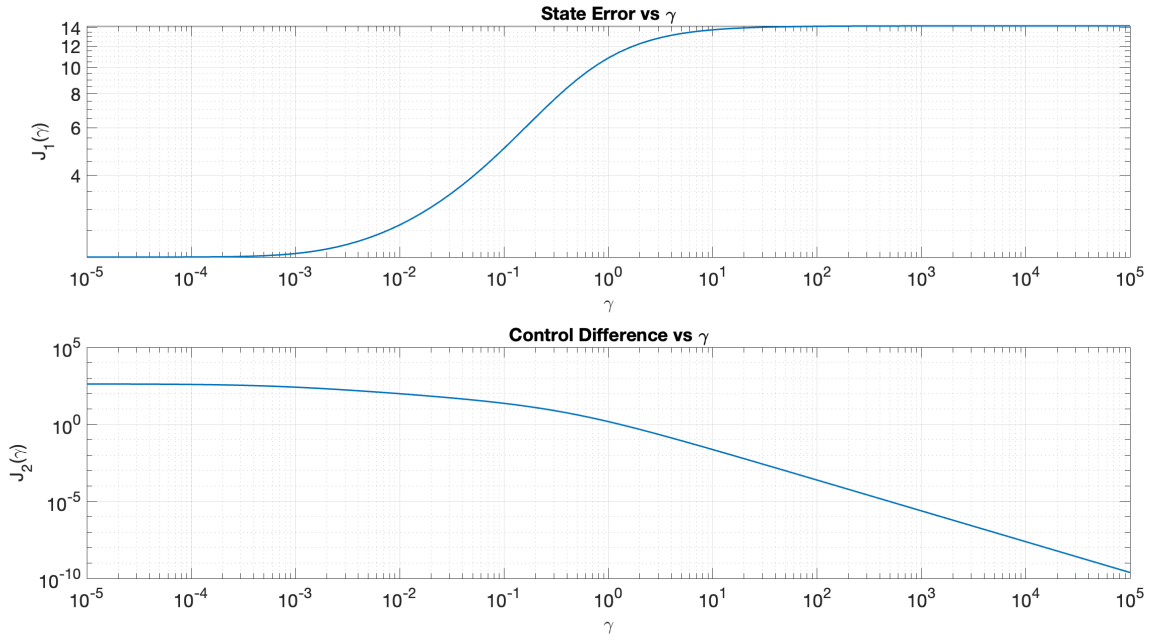


Figure 9: Functionals $J_1(\gamma)$ (top) and $J_2(\gamma)$ (bottom) versus the regularization parameter γ.

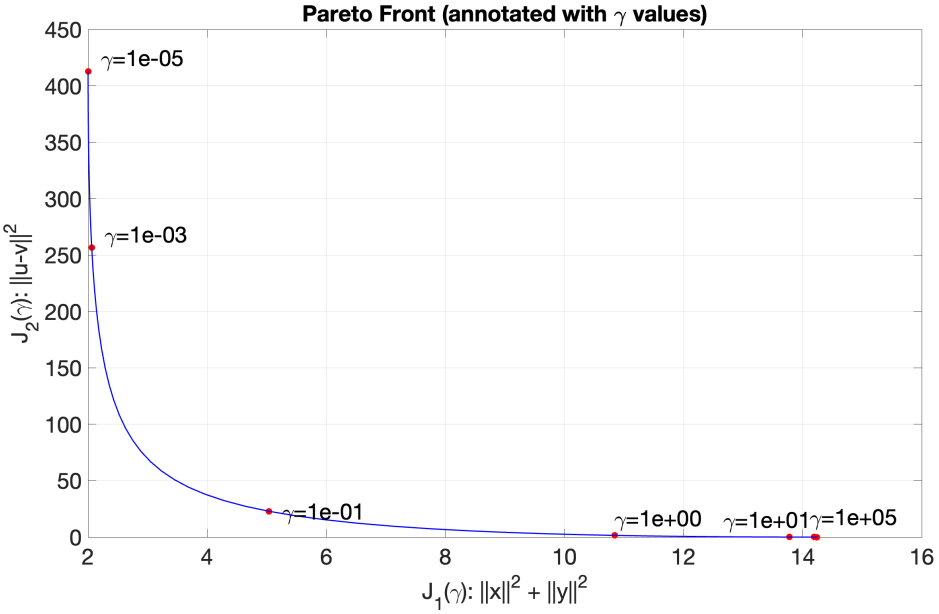


Figure 10: Pareto front showing the trade-off between state error J_1 and control difference J_2. Each point corresponds to a different value of γ. From here, we observe that a significant decrease of the total cost (dominated by J_2 can be achieved with a small increase of γ until approximately $\gamma < 0.1$, after which further increasing γ has no relevant effect.

0.1 Part II.d [4 marks]

The Huber-like regularizer $L_\epsilon : \mathbb{R} \to \mathbb{R}$ is defined as:

$$L_\epsilon(z) = \begin{cases} \frac{1}{2}z^2 & \text{if } |z| \leq \epsilon \\ \epsilon\left(|z| - \frac{1}{2}\epsilon\right) & \text{if } |z| > \epsilon \end{cases}$$

This is a smooth approximation of $|z|$, with a similar behavior when $\epsilon \to 0$. We check its smoothness:

$$\frac{dL_\epsilon}{dz}(z) = \begin{cases} z & \text{if } |z| \leq \epsilon \\ \epsilon \cdot \text{sign}(z) & \text{if } |z| > \epsilon, \end{cases}$$

which is continuous at $z = \pm\epsilon$.

Gradient Descent Implementation

The objective function is:

$$f(u, v) = \|B_x u + c_x\|_2^2 + \|B_y v + c_y\|_2^2 + \gamma_2\|u - v\|_2^2 + \gamma_1 \sum_{i=1}^{N} L_\epsilon(u_i - v_i).$$

Because of the regularization term, we can no longer solve this as a LLS problem. We must use gradient descent. The gradient with respect to $w = (u, v)$ is:

$$\frac{\partial f}{\partial u_i} = 2(B_x^\top (B_x u + c_x))_i + 2\gamma_2(u_i - v_i) + \gamma_1 \frac{dL_\epsilon}{dz}(u_i - v_i)$$
$$\frac{\partial f}{\partial v_i} = 2(B_y^\top (B_y v + c_y))_i - 2\gamma_2(u_i - v_i) - \gamma_1 \frac{dL_\epsilon}{dz}(u_i - v_i).$$

We implement gradient descent with constant stepsize, and the following settings:

- Step size of $\alpha = 0.01$ (constant).
- Maximum iterations Varies by case (100,000 to 1,000,000)
- Convergence criterion is determined by $\|\nabla f\| < 10^{-8}$ or maximum iterations reached. In all our tests, the gradient tolerance has been reached before the maximum number of iterations.
- The method is initialized with $w^{(0)} = 0$.

Numerical Results

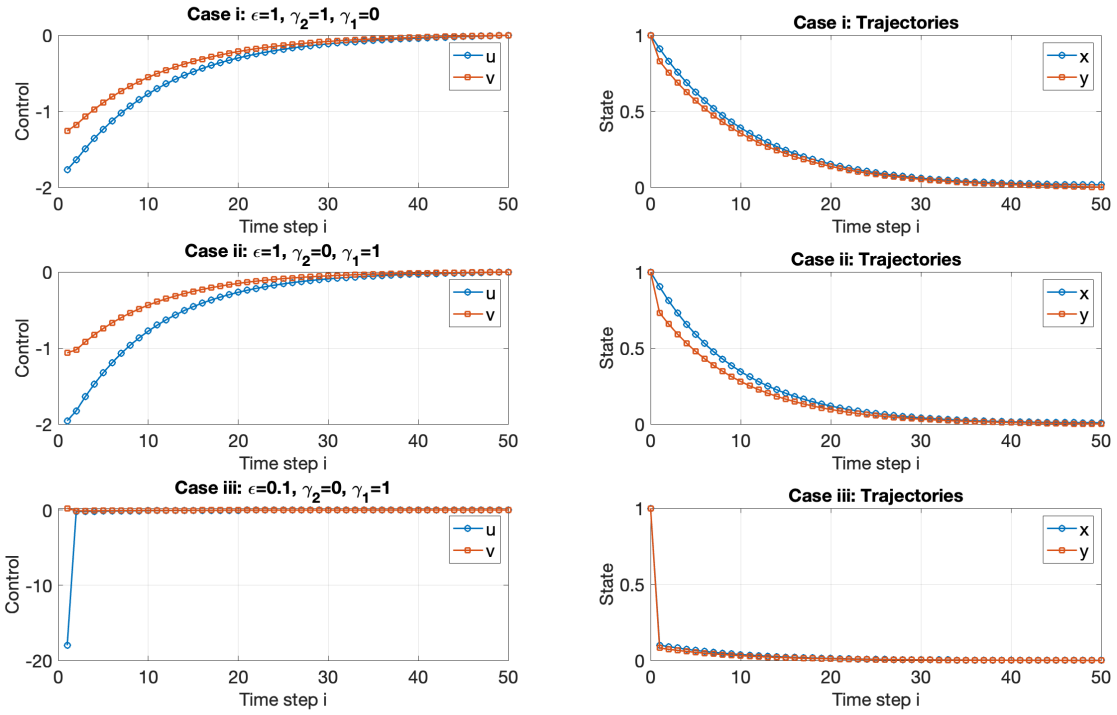


Figure 11: Comparison of controls (left column) and trajectories (right column) for three cases. The top row is consistent with part II.b, the main difference between middle and bottom rows comes from the smaller ϵ regularization. A larger ϵ value makes the regularizer behave like an ℓ_2 penalty, hence similar to to problem in part II.b. As ϵ is decreased, the regularizer mimics the ℓ_1 norm, promoting sparsity. This is observed in the fast convergence of the control signals to 0. Indirectly, this affect the stabilization of the trajectories.

General feedback

Overall, the submission level was excellent. For Part I, most submissions correctly identified the interplay between data, polynomial degree (parameters), and the impact of gradient-enhanced data. A number of submissions used linear plots which prevent a meaningful analysis, as error plots look flat. Semi-log plots are much more desirable in this case.

For part II, submissions were of very good level. Many submissions did not include a quantitative analysis of the Pareto front in c); it is necessary to understand the scaling of γ and its impact on both functionals. For part d), a number of submissions found close-to but still suboptimal results. This is due to setting gradient descent with a very small time step, no gradient norm verification, or insufficient number of iterations.